

NONCLINICAL REPORT

Nonclinical Report No. VRC04/NHP01

Immunogenicity and Protective Efficacy of mRNA-1273 in Rhesus Macaques

SPONSOR:

Vaccine Research Center
National Institute of Allergy and Infectious Diseases
National Institutes of Health
Bethesda, MD 20892

PERFORMING LABORATORY:

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DATE

12 October 2020

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ACE-2	angiotensin-converting enzyme 2
ATCC	American Type Culture Collection
AUC	area under the curve
BAL	bronchoalveolar lavage
BSA	bovine serum albumin
cISH	chromogenic in situ hybridization
CoV	coronavirus
DAB	3,3' diaminobenzidine
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
ECL	electrochemiluminescence
ELISA	enzyme-linked immunosorbent assay
GMP	Good Manufacturing Practice
GMT	geometric mean titer
H&E	hematoxylin and eosin
IACUC	Institutional Animal Care and Use Committee
ID ₅₀	50% inhibitory dilution
Ig	immunoglobulin
IHC	immunohistochemical
IL	interleukin
IM	intramuscular
IN	intranasal(ly)
IT	intratracheal(ly)
LNP	lipid nanoparticle
MIMOSA	Mixture Models for Single-Cell Assays
mRNA	messenger RNA
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NIAID	National Institute of Allergy and Infectious Disease
NIH	National Institutes of Health
NS	nasal swab(s)
PBS	phosphate-buffered saline
PCR	polymerase chain reaction
PEG	polyethylene glycol
PEG2000-DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
PFU	plaque-forming units
RBD	receptor binding domain
RT-qPCR	reverse transcription quantitative PCR
S	spike
S1 NTD	N-terminal domain of S1
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS-CoV-2	2019 novel coronavirus

Abbreviation	Definition
sgRNA	subgenomic RNA
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate
Tfh	T-follicular helper
Th1	T-helper 1
Th2	T-helper 2
Tris HCl	tris(hydroxymethyl)aminomethane hydrochloride
USAMRIID	United States Army Medical Research Institute of Infectious Disease
UTR	untranslated region
VRC	Vaccine Research Center

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that the 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized spike protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

The objectives of Study VRC04 were to evaluate the immunogenicity of mRNA-1273 and protection from SARS-CoV-2 challenge in rhesus macaques.

ModernaTX, Inc. is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to ModernaTX, Inc.'s mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively. The test articles in this study consisted of mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP) and phosphate-buffered saline (PBS).

Rhesus macaques (n=8/group) were vaccinated with 10 µg or 100 µg of mRNA-1273 or PBS via intramuscular (IM) injection on a prime (Week 0)/boost (Week 4) schedule. As a negative control, PBS was administered. At Study Week 8, rhesus macaques were challenged intratracheally (IT) and intranasally (IN) with a total dose of 7.6×10^5 plaque-forming units (PFU) of SARS-CoV-2 isolate USA-WA1/2020. Pre-challenge serum samples were collected Week -1, Week 0, Week 2, Week 4, Week 6, and Week 8 for serology and T-cell analysis. Post-challenge serum samples were collected on Day 2, Day 4, and Day 7, nasal swab (NS) samples were collected on Day 1, Day 2, Day 4, and Day 7, and bronchoalveolar lavage (BAL) samples were collected on Day 2, Day 4, and Day 7. Lung sections were collected for histopathological analysis and viral quantification on Day 7 or 8, and Day 14 or 15 post-challenge.

Antibody and T-cell responses were assessed pre- and post-challenge, and protection after challenge was assessed in the upper and lower airways from NS samples and BAL fluid by polymerase chain reaction (PCR) to measure viral RNA. Reverse-transcription quantitative PCR (RT-qPCR) using primers specific for subgenomic RNAs (sgRNAs) was used to quantify nascent

viral replication. Assessments of pathology and viral quantification were performed on lung tissue specimens.

Dose-dependent increases in binding antibody and neutralizing activity were observed in rhesus macaques vaccinated with mRNA-1273, with increasing response measured from the boosting dose. mRNA-1273 elicits S receptor binding domain (RBD) subdomain and the N-terminal domain of S1 (S1_NTD) subdomain-specific immunoglobulin (Ig) G titers and induced T-helper 1 (Th1)-directed CD4 T-cell and interleukin (IL)-21-producing T-follicular helper (Tfh)-cell responses with no evidence of a T-helper 2 (Th2)-directed response. In addition, mRNA-1273 drives potent neutralizing antibody responses as measured in both live virus and pseudovirus neutralization assays, with higher titers measured at the 100 µg dose level. Post-challenge, evaluation of the protective efficacy against upper and lower SARS-CoV-2 infection revealed a lack of detectable sgRNA in BAL fluid and NS samples in animals dosed at 100 µg and in BAL fluid of animals dosed at 10 µg as soon as 2 days post-challenge. Additionally, RNA copies (measured by PCR) showed a decrease in virus in BAL fluid in the 10 and 100 µg group and NS samples in the 100 µg group 2 days post-challenge. There was limited inflammatory cytokine induction in upper and lower airways of rhesus macaques within 2 days of viral challenge, suggesting that the rapid control of the virus was sufficient to limit innate immune activation. There was also a dose-dependent increase in S-specific IgG in BAL fluid compared with the levels in control animals, with no increase in titers post-challenge. This is indicative of rapid control of the virus without induction of an anamnestic response to the virus. Finally, limited (10 µg-dosed animals) to no (100 µg-dosed animals) lung inflammation was measured post-challenge, whereas moderate to severe inflammation was measured in the lungs of control animals. No evidence of vaccine-associated immunopathology was found in postmortem tissue analysis.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

Vaccine Research Center (VRC)
National Institute of Allergy and Infectious Diseases (NIAID)
National Institutes of Health (NIH)
Bethesda, MD 20892

2.1.2. Performing Laboratory

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2.1.3. Study Number

Study number VRC-20-857.1 (VRC04).

2.1.4. Purpose of the Study

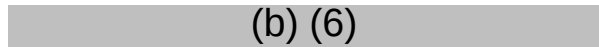
The objectives of this study were to evaluate the immunogenicity of mRNA-1273 and protection from SARS-CoV-2 challenge in rhesus macaques.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved and conducted in accordance with the regulations of the ModernaTX, Inc. Institutional Animal Care and Use Committee ([IACUC] Protocol VRC-20-857).

2.1.6. Study Director

(b) (6)



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2.1.7. Key Personnel

Role	Study VRC04
General coordinator	Vaccine Research Center National Institute of Allergy and Infectious Diseases National Institutes of Health Bethesda, MD 20892
Study coordinator	(b) (4), (b) (6)
Analytical assays: ACE-2 Binding inhibition lead	McDermott Lab Vaccine Immunology Program 9 West Watkins Mill Road Gaithersburg, MD 20878
Analytical assays: Nanoluc Assay lead	(b) (4), (b) (6)
Report compilation	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Performance

Schedule of Events	Study VRC04
Test article administration	Week 0 (prime): 9 April 2020 Week 4 (boost): 6 May 2020
Challenge with SARS-CoV-2	Week 8: 8 June 2020
Pre-challenge samples collected	Blood (Week -1, Week 0, Week 2, Week 4, Week 6, and Week 8)

Post-challenge samples collected	Blood (Day 2, Day 4, and Day 7) Nasal Swabs (Day 1, Day 2, Day 4, and Day 7) Bronchoalveolar lavage (Day 2, Day 4, and Day 7) Lung sections (Day 7 or Day 8, and Day 14 or Day 15)
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2.1.9. Records Maintained

Data are archived in the Viral Pathogenesis Laboratory (VRC) database, which is password protected and can be accessed only by authorized personnel. The server is maintained by NIH information technology staff and is backed up remotely on a daily basis.

2.2. Test Articles

2.2.1. mRNA-1273 Test Article

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102; cholesterol; DSPC; and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in this study was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with the same method as the Good Manufacturing Practice (GMP) mRNA-1273 clinical Drug Product ([Appendix 1](#)) This product is a novel LNP-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 was manufactured by ModernaTX, Inc. A sequence-optimized mRNA that encodes prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine ([Nelson et al 2020](#)). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O-methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4) buffer exchanged (b) (4), sterile filtered, and kept frozen at -20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described (Hassett et al 2019) (b) (4)

(b) (4) The mixture was neutralized with tris(hydroxymethyl)aminomethane hydrochloride (Tris HCl), pH 7.5; sucrose was added as a cryoprotectant; and the final solution was sterile filtered. Vials were filled with mRNA-1273-formulated LNP and stored frozen at -70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin, and bioburden, and the material was deemed acceptable for the in vivo study (Corbett et al 2020).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No	mRNA Lot No.	Size (nm)	PDI	EE%
mRNA-1273 Drug Product	mRNA encoding SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; S = spike; SARS-CoV-2 = 2019 novel coronavirus; SM-102 = heptadecan-9-yl 8-((2-hydroxyethyl) (6-oxo-6-(undecyloxy)hexyl) amino)octanoate; Tris HCl = tris(hydroxymethyl)aminomethane hydrochloride.

Note: All mRNA test material stocks were formulated in SM-102-containing LNPs with a final storage buffer in 10.7 mM sodium acetate, 8.7% sucrose, and 20 mM Tris HCl at pH 7.5 and stored at -70°C .

The EE% is the percentage of mRNA (b) (4)

(b) (4)

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4)) was purchased commercially.

2.3. Test System

Experiments were conducted in 24 Indian-origin rhesus macaques (12 female and 12 male; 3 to 6 years old) provided by the VRC and transferred to (b) (4) After SARS-CoV-2 infection, rhesus macaques have transient viral replication in the upper and lower airways and mild to moderate inflammation in the lung that resolves within 14 days (Yu et al 2020). Thus, rhesus macaques are a useful animal model for assessing vaccine-mediated protection against early viral replication.

Human convalescent serum was obtained from adults with mild, moderate, or severe COVID-19 and a history of laboratory-confirmed SARS-CoV-2 infection. Details are provided in [Section 2.4.3](#).

2.4. Methods

2.4.1. Study Design and Treatment Groups

Indian-origin rhesus macaques (male [n=12] and female [n=12]) were sorted according to sex, age, and weight and randomly assigned to 1 of 3 treatment groups according to the study design presented in [Table 2](#). Rhesus macaques (n=8/group) were vaccinated with 10 µg or 100 µg of mRNA-1273 or PBS via IM injection in the right hind leg on a prime (Week 0)/boost (Week 4) schedule. Four weeks post-boost, all animals were challenged with a total dose of 7.6×10^5 PFU from a stock of 1.9×10^5 PFU/mL SARS-CoV-2 (USA-WA1/2020 strain). The challenge was administered in a total volume of 4 mL (3 mL IT and 1 mL IN [0.5 mL per nostril]). Antibody and T-cell responses were assessed pre- and post-challenge. Bronchoalveolar lavage fluid was obtained on Day 2, Day 4, and Day 7 post-challenge, and NS specimens were obtained on Day 1, Day 2, Day 4, and Day 7 post-challenge; active viral replication and viral genomes were assessed by PCR. Histopathological analysis and viral quantification were performed on lung tissue specimens collected after animals were euthanized on Day 7 or Day 8, and Day 14 or Day 15 post-challenge.

Table 2: Treatment Regimen for Rhesus Macaques

Group (n=8)	Treatment	Vaccination Schedule		Immunization Phase Endpoints	Challenge Week 8 ^a	Post-challenge Endpoints
		Prime (Week 0)	Boost (Week 4)			
1	PBS	NA	NA	Binding titers by ELISA Neutralizing activity by pseudovirus and live-virus neutralization assays Th1 vs Th2 T-cell responses by ICS ACE-2 binding inhibition	WT SARS-CoV-2	Daily clinical evaluation Viral load by RT-qPCR (nasal swabs and BAL fluid) Lung pathology Plasma/serum cytokines
2	mRNA-1273	10 µg	10 µg			
3	mRNA-1273	100 µg	100 µg			

Abbreviations: ACE-2 = angiotensin converting enzyme-2; BAL = bronchoalveolar lavage; ELISA = enzyme linked immunosorbent assay; ICS = intracellular cytokine staining; NA = not applicable, PBS = phosphate-buffered saline; PFU = plaque-forming units; RT-qPCR = reverse transcription quantitative polymerase chain reaction; SARS-CoV-2 = 2019 novel coronavirus; Th1 = T helper 1; Th2 = T helper 2.

^a Animals were challenged with a total dose of 7.6×10^5 PFU from a stock of 1.9×10^5 PFU/mL SARS-CoV-2 (USAWA1/2020). The challenge was administered in a total volume of 4 mL (3 mL intratracheal and 1 mL intranasal [0.5 mL per nostril]) of SARS-CoV-2.

2.4.2. Antibody Measurements

2.4.2.1. Serum Antibody Measurements by Enzyme-linked Immunosorbent Assay

SARS-CoV-2 S-2P, RBD, and S1_NTD-specific IgG antibody titers in sera were determined by enzyme-linked immunosorbent assay (ELISA), similar to previously published methods (Corbett et al 2020). Briefly, (b) (4) plates (b) (4) were coated with SARS-CoV-2 S-2P (100 ng/well) in 1X PBS at 4°C for 16 hours. After standard washes and blocks, plates were incubated with duplicate serial dilutions (1:100, 4-fold, 8X) of heat-inactivated sera for 1 hour at room temperature. Following washes, anti-monkey IgG (H+L) secondary antibody ((b) (4)) conjugated in horseradish peroxidase in a blocking buffer at a 1:10,000 dilution ((b) (4)) was added. 3,5,3'-Tetramethylbenzidine ((b) (4)) substrate was used to detect antibody responses. The optical density of each well was read at 450 nm and 650 nm by a (b) (4) (b) (4). Area under the curve (AUC) was calculated using Prism v8 (GraphPad), setting the baseline with a defined endpoint (average of negative control wells + 10STDEV) and taking the total peak area. Area under the curve indicates the amount of IgG binding to S-2P over time. A MULTI-ARRAY ELISA was used to detect RBD and S1_NTD-specific IgG antibodies in sera (Section 2.4.2.2).

2.4.2.2. Measurements of Antibody Isotype Responses in Bronchiolar Lavage Fluid

Total IgG and IgA antigen-specific antibodies were determined by ELISA using MULTI-ARRAY 384-well streptavidin-coated plates ((b) (4))). Plates were blocked with 5% bovine serum albumin (BSA) in 1X PBS for 30 minutes at room temperature. After blocking, plates were washed with wash buffer (0.05% Tween-20 in 1X PBS). Plates were coated with 1 µg/mL SARS-CoV-2 S-2P-Foldon-AVI-biotin protein, 0.18 µg/mL SARS-CoV-2 RBD-AVI-biotin protein, or 0.26 µg/mL SARS-CoV-2 NTD-AVI-biotin protein (Zhou et al 2020) in 1% BSA in 1X PBS for 1 hour at room temperature. After incubation, plates were washed. Serial dilutions of 10X concentrated BAL fluid in 1% BSA, 0.05% Tween-20 in 1X PBS were added to wells and incubated for 1 hour at room temperature. Plates were washed 5X with wash buffer and incubated for 1 hour with 1 µg/mL of anti-human IgG sulfo-tag or 2 µg/mL of anti-human/nonhuman primate IgA sulfo-tag ((b) (4)) in 1% BSA, 0.05% Tween 20 in 1X PBS. Plates were then washed 5X with wash buffer, and 1X Read T buffer was added to each well. Signals were read by ((b) (4)) plate reader ((b) (4)).

2.4.2.3. ACE-2 Binding Inhibition Assay

((b) (4)) 384-well 4-Spot Custom Serology ((b) (4)) plates precoated with RBD were blocked for 30 minutes with ((b) (4)) Blocker A (5% BSA) at room temperature. The blocking solution was washed off (3 cycles of PBS-Tween), and test samples were applied in duplicate to

the wells at 1:40 dilution and allowed to incubate with shaking (1300 rpm) for 1 hour at room temperature. (b) (4) Assay Diluent 100 in quadruplicate was used as the negative control. (b) (4) labeled angiotensin-converting enzyme 2 (ACE-2; supplied at 200X) was applied to the wells and allowed to associate with the test samples and RBD within the assay wells for 1 hour at room temperature with shaking (1300 rpm). Plates were then washed (3 cycles of PBS-Tween) to remove unbound detection antibody. (b) (4) read buffer substrate was applied to the wells, and the plate was entered into the (b) (4) detection system.

A current was applied to the plate, and areas of well surface where RBD had complexed with (b) (4) -ACE-2 emitted light in the presence of the electrochemiluminescence (ECL) substrate. The (b) (4) detection system quantitated the amount of light emitted and reported the ECL unit response as a result for each test sample and control of the plate. The amount of signal emitted in wells containing no sample (assay diluent only) was evaluated as the maximal binding response. Reduction of ECL response from this maximal readout was directly proportional to the extent of competitive binding activity in the test sample.

2.4.2.4. Pseudovirus Neutralization Assay

(b) (4)

(b) (4)

, and neutralization was assessed as previously described. y, triplicate heat-inactivated sera was mixed with pseudoviruses, incubated, and then added to ACE-2-expressing 293T-cells. Seventy-two hours later, cells were lysed and luciferase activity (relative light units) was measured. Percent neutralization was normalized considering uninfected cells as 100% neutralization and cells infected with only pseudovirus as 0% neutralization. Fifty percent inhibitory dilution (ID₅₀) titers were determined using a log (inhibitor) versus normalized response (variable slope) nonlinear function in Prism v8 (GraphPad).

2.4.2.5. NanoLuc Virus Reporter Assay

Four weeks after the second vaccination, immediately before challenge, live-virus neutralization was assessed by NanoLuc reporter assay. A full-length SARS-CoV-2 virus based on the Seattle Washington isolate was designed to express luciferase and was recovered via reverse genetics (Yount et al 2003; Scobey et al 2013; Hou et al 2020). Virus titration in Vero E6 cells from ATCC that had been in culture at the United States Army Medical Research Institute of Infectious Disease (USAMRIID) were used to obtain a relative light units signal of at least 10X the cell-only control background. Vero E6 USAMRIID cells were plated at 20,000 cells per well in

clear bottom black walled 96-well plates. On the next day, neutralizing antibody serum samples were tested at a starting dilution of 1:50 and were serially diluted 2-fold up to 10 dilutions. Antibody-virus complexes were incubated at 37°C with 5% carbon dioxide for 1 hour. Following incubation, the media was removed, and virus-antibody dilution complexes were added to the cells in duplicate. Virus-only controls and cell-only controls were included in each neutralization assay plate. Following infection, plates were incubated at 37°C with 5% carbon dioxide for 24 hours. After the 24-hour incubation, cells were lysed and luciferase activity was measured via (b) (4) Luciferase Assay System (b) (4), according to manufacturer specifications. SARS-CoV-2 neutralization titers were defined as the sample dilution at which a 50% reduction in relative light units was observed relative to the average of the virus control wells.

2.4.3. Human Convalescent-Phase Serum

Human convalescent-phase sera (n=42) were obtained from adults between 18 and 84 years old with mild, moderate, or severe COVID-19 and a history of laboratory-confirmed SARS-CoV-2 infection approximately 1 to 2 months before providing specimens. These specimens were obtained under institutional review board-approved sample collection protocols at the National Institute of Health Clinical Center, Bethesda, MD (NCT00067054), Aaron Diamond AIDS Research Center, Columbia University, New York, NY (NCT04342195), and the University of Washington, Seattle, WA (HAARVI study and STUDY00000959). Written informed consent was provided by the participants.

2.4.4. Quantification of SARS-CoV-2 RNA in Bronchiolar Lavage Fluid and Nasal Swab Samples

SARS-CoV-2 in the lungs was quantified by plaque assay. Polymerase chain reaction was used to determine viral RNA load in BAL fluid and NS samples; both methods are previously described (Yu et al 2020). To quantify replicating virus in BAL fluid and NS samples, sgRNA primers were utilized, as described in previously published methods (Wölfel et al 2020).

2.4.5. Measurement of Cytokines and Chemokines in BAL Fluid

Bronchoalveolar fluid was analyzed with (b) (4). The BAL fluid was concentrated (10X) and added neat in duplicate to (b) (4) Nonhuman Primate Cytokine Magnetic Bead Panel (b) (4) and the standard assay was performed according to manufacturer's directions. Fluorescence data was collected on (b) (4) with (b) (4).

2.4.6. Intracellular Cytokine Staining

Cryopreserved peripheral blood mononuclear cells were thawed and rested overnight at 37°C in 5% carbon dioxide. Cells were then stimulated with SARS-CoV-2 S protein (S1 and S2,

homologous to the vaccine insert) and nucleoprotein peptide pools (b) (4), (b) (4) at a final concentration of 2 µg/mL in the presence of monensin and co-stimulatory antibodies anti-CD28 and anti-CD49d (b) (4) for 6 hours. Negative controls received an equal concentration of dimethyl sulfoxide (instead of peptides) and co-stimulatory antibodies. Intracellular cytokine staining was performed as described (Donaldson et al 2019) except the following monoclonal antibodies were added: PD-1 BUV737 (b) (4) in place of PD-1 BV785, TNF-FITC (b) (4) in place of IL-5 BB515, and CD40L BV785 (b) (4). (b) (4) was used to exclude dead cells. All antibodies were previously titrated to determine the optimal concentration. Samples were acquired on an (b) (4) flow cytometer and analyzed using FlowJo version 9.9.6 (Treestar, Inc.).

2.4.7. Histopathology

Lung tissue from SARS-CoV-2-infected rhesus macaques was processed per a standard protocol. Briefly, the tissues were fixed in 10% neutral buffered formalin, processed with a (b) (4) processor (b) (4), embedded in paraffin, and sectioned at 5 µm for histological analysis. Tissue sections were stained with hematoxylin and eosin (H&E) for routine histopathology.

2.4.7.1. Immunohistochemistry

Immunohistochemical (IHC) staining was performed using formalin-fixed paraffin-embedded tissue sections (5 µm) obtained from lung tissues. Immunohistochemical staining used a rabbit polyclonal SARS-CoV-2 (b) (4) at a dilution of 1:2000. Staining was carried out on the (b) (4) platform according to manufacturer-supplied protocols. Briefly, 5-µm-thick sections were deparaffinized and rehydrated. Heat-induced epitope retrieval was performed using (b) (4), pH 6.0, heated to 100°C for 20 minutes. Specimens were then incubated with hydrogen peroxide to quench endogenous peroxidase activity prior to applying the primary antibody. Detection with 3,3' diaminobenzidine (DAB) chromogen was completed using a (b) (4) (b) (4). Slides were finally cleared through gradient alcohol and xylene washes prior to mounting and placing cover slips.

All tissue sections were examined by a board-certified veterinary pathologist using an (b) (4) (b) (4) light microscope, and photomicrographs were taken using an (b) (4) camera.

2.4.7.2. In Situ Hybridization

Chromogenic in situ hybridization (cISH) was performed using RNAScope technology. The target probe detecting positive-sense SARS-CoV-2 RNA (b) (4)

(b) (4) was designed by (b) (4). Epitope retrieval was performed on tissues with an application of (b) (4) for 15 minutes at 95°C followed by a 15-minute digestion with (b) (4) at 40°C. Tissues were hybridized with the probes for 2 hours and visualized with the (b) (4) according to the manufacturer's protocol.

2.5. Data Presentation and Analysis

The primary endpoints of the study were reduction of viral load in the BAL fluid of the mRNA-1273 groups compared to the PBS control group. The efficacy endpoint analysis used RT-qPCR and sgRNA RT-qPCR, similar to other nonhuman primate SARS-CoV2 vaccine studies ([Munster et al 2020](#); [van Doremalen et al 2020](#)) to quantify viral RNA and replicating virus, respectively. In the BAL fluid and NS specimens, the prespecified primary analysis measured peak viral load over Days 2 through 7 for each animal. Outputs were compared with Mann-Whitney tests and a hierarchical testing procedure (in which the 100 µg group was compared with the control group first, and then the 10 µg group was compared to the 100 µg group and the control group if the first comparison had a $p < .05$).

Statistical analyses were performed in R version 4.0. For all other measures, groups were compared by Kruskal-Wallis test, followed by pairwise Mann-Whitney tests with Holm's adjustment, on the set of pairwise tests if the Kruskal-Wallis test indicated significance. For estimation and testing of correlations, Spearman's nonparametric method was used. Intracellular cytokine response rates were determined by implementing the Mixture Models for Single-Cell Assays (MIMOSA) algorithm. ([Finak et al 2013](#)).

Graphs were generated using Prism v8 (GraphPad Software). Unless otherwise noted, all graphs are represented as box and whisker plots depicting the median by a horizontal line, the top and bottom of the box in the interquartile range, and the whiskers range. ICS response rates were estimated using the MIMOSA package, which uses a hierarchical beta-binomial model to calculate the probability that each individual is responding to a specific stimulation; values that were judged to be a probable response, subject to a 0.05 false discovery rate constraint, are considered responses for these analyses.

3. RESULTS

3.1. Binding Antibody Responses and Neutralizing Activity After mRNA-1273 Vaccination

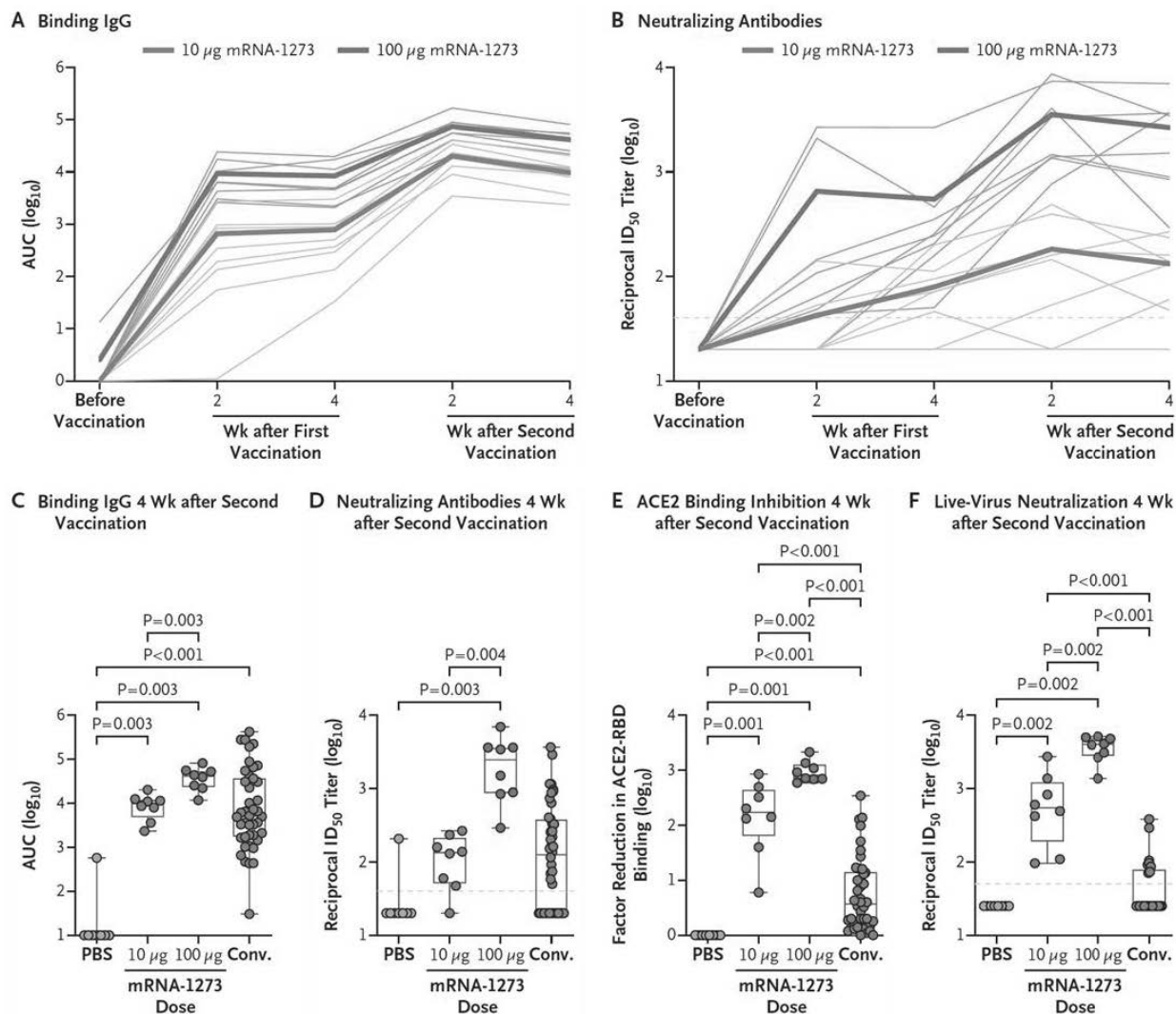
Antibody neutralization was measured using both a live-virus assay (NanoLuc Virus Reporter Assay, [Section 2.4.2.5](#)) and a pseudovirus assay ([Section 2.4.2.4](#)). Both assays are commonly used to assess neutralizing activity in nonclinical studies for a variety of vaccine candidates.

Four weeks post-boost and immediately prior to viral challenge, IgG binding titers to S-2P increased over baseline in a dose-dependent manner, reaching AUC values of 8241 and 36,186 for the 10 and 100 µg mRNA-1273 groups, respectively ([Figure 1A](#)). Similarly, there was a dose-dependent increase in neutralizing activity. Four weeks post-prime, animals immunized with 10 µg of mRNA-1273 had reciprocal 50% inhibitory dilution (ID₅₀) geometric mean titer (GMT) of 63, which increased to 103 by 4 weeks post-boost. Neutralizing activity in animals administered 100 µg of mRNA-1273 at 4 weeks post-prime demonstrated an ID₅₀ GMT of 305, 5 times that at the 10 µg-dose, which rose to 1862 post-boost ([Figure 1B](#)). Of note, S-specific IgG binding ([Figure 1C](#)) and neutralizing GMT ([Figure 1D](#)) observed 4 weeks post-boost in the 100 µg group were 5- and 15-fold higher, respectively, than the mean of convalescent human sera (n=42). Neutralizing activity was measured using a live SARS-CoV-2 reporter virus, and 4 weeks post-boost, rhesus macaques immunized with 10 and 100 µg of mRNA-1273 had reciprocal ID₅₀ GMTs of 501 and 3481, respectively ([Figure 1F](#)).

These analyses were extended to assess antibody responses to specific domains of the S protein (RBD and S1_NTD) and to measure functional viral inhibition.

The results indicated that mRNA-1273 elicited more potent RBD- and S1_NTD-specific serum antibody responses than that observed in convalescent human sera ([Figure 2A](#) and [Figure 2B](#), respectively). Also, inhibition of ACE-2 binding to RBD in the 100 µg group was 938-fold higher than inhibition in the PBS control group and 348-fold higher than inhibition from convalescent human sera ([Figure 1E](#)).

Taken together, these data show that mRNA-1273 targets both RBD and S1_NTD subdomains with potent neutralizing capacity and induces robust S-specific antibody responses.

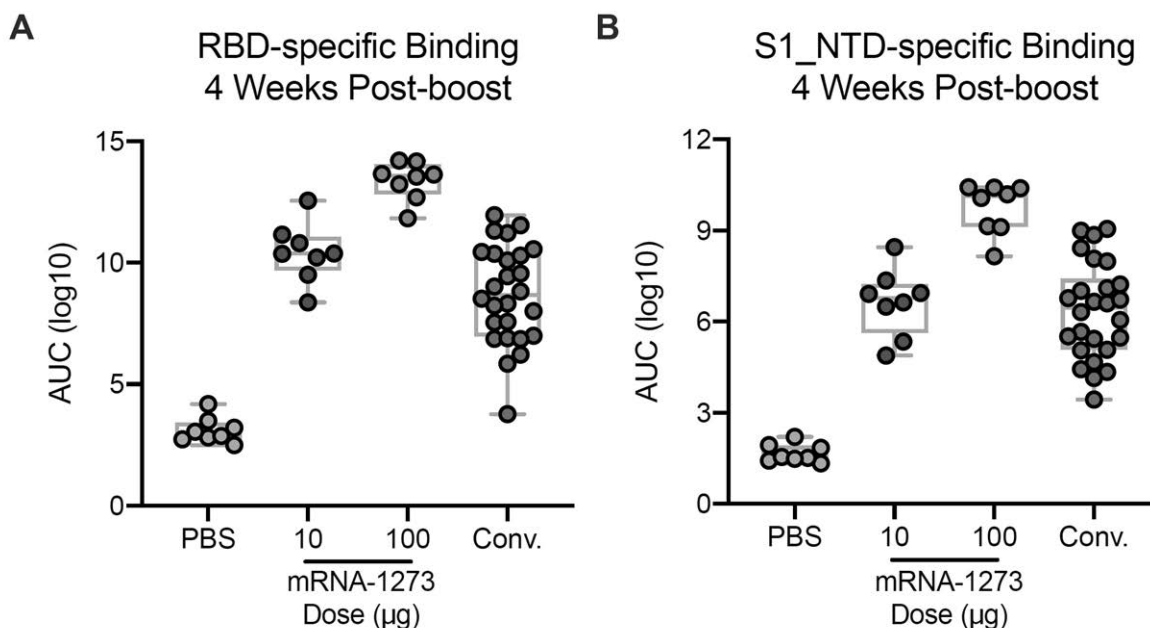
Figure 1: Antibody Responses After mRNA-1273 Immunization in Rhesus Macaques

Abbreviations: ACE-2 = angiotensin-converting enzyme 2; AUC = area under the curve; Conv, convalescent human serum; ID₅₀ = 50% inhibitory dilution; Ig = immunoglobulin; PBS = phosphate-buffered saline; Wk = week(s).

Note: In Panels A and B, faint lines represent individual animals and bold lines represent the geometric mean titer for each group. In Panels C, D, E, and F, results were compared with the antibody responses in a panel of convalescent-phase human serum specimens (42 specimens in Panels C, D, and E and 26 specimens in Panel F).

Source: [Corbett et al 2020, Figure 1](#).

Figure 2: SARS-CoV-2 Spike Antibody Domain Specificity Following mRNA-1273 Immunization in Rhesus Macaques



Abbreviations: AUC = area under the curve; Conv = convalescent human sera; PBS = phosphate-buffered saline; RBD = receptor binding domain; S1_NTD = N-terminal domain of S1.

Source: [Corbett et al 2020, Supplemental Figure 2](#).

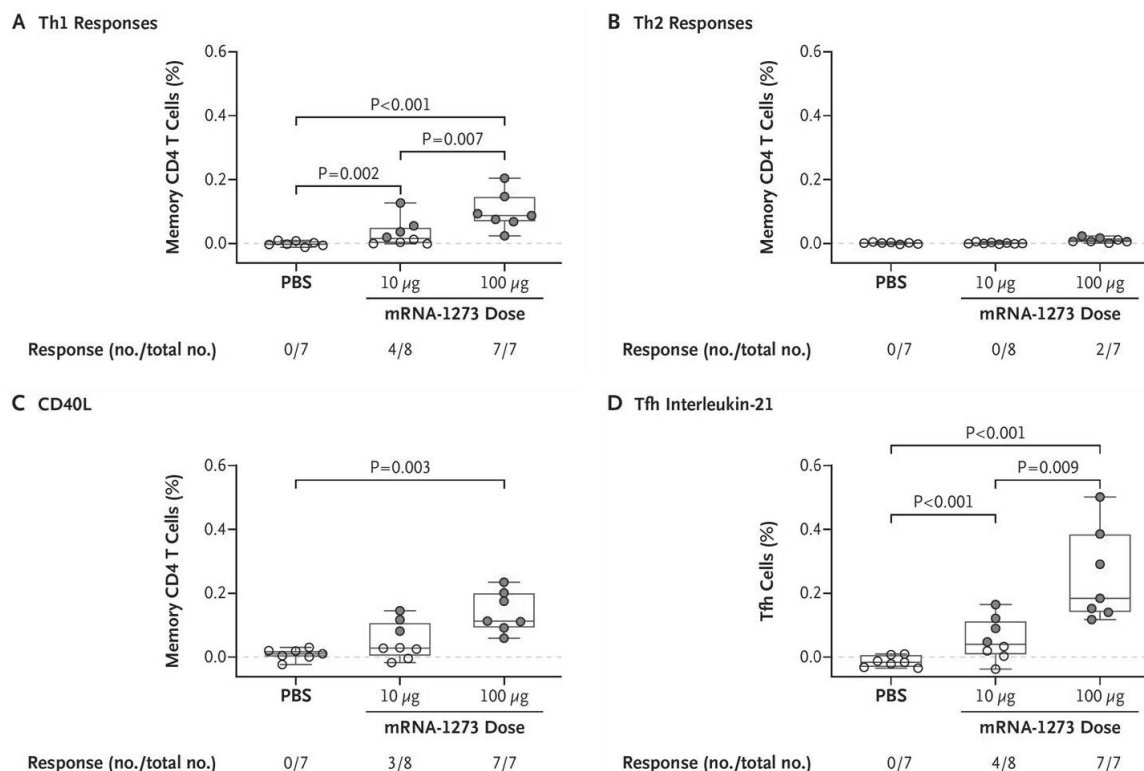
3.2. T-Cell Responses Following mRNA-1273 Vaccination

Studies have shown that SARS-CoV-2-specific T-cell immunity may have a role in pathogenesis or protection against SARS-CoV-2 and may influence the humoral immune response ([Meckkiff et al 2020](#); [Neidleman et al 2020](#); [Weiskopf et al 2020](#)). Activated CD4 T-cells can be segregated into Th1- and Th2-directed responses based on the production of specific cytokines and are critical for B cell activation and antibody production. Importantly, the induction of CD4 Th2 T-cell responses (production of IL-4, IL-5 or IL-13) has been linked to vaccine-associated enhanced respiratory disease, as seen in some human respiratory virus infections (eg, respiratory syncytial virus, measles [[Wang et al 2012](#); [Hou et al 2020](#)] and in animal models of SARS-CoV infection [[Scobey et al 2013](#)]); however, vaccine-associated enhanced respiratory disease has not been observed with exclusive CD4 Th1 T-cell responses (production of interferon- γ , IL-2, tumor necrosis factor- α) ([Grifoni et al 2020](#); [Peng et al 2020](#); [Sekine et al 2020](#); [Weiskopf et al 2020](#)).

Intracellular cytokine staining was performed on peripheral blood mononuclear cells collected 4 weeks post-boost, immediately before viral challenge, to assess T-cell responses to the SARS-CoV-2 S1 peptide pool. A 19-color multiparameter flow cytometry was used to assess the functional heterogeneity of S-specific CD4 and CD8 T-cell cytokine responses. Four weeks

post-boost, a dose-dependent increase in Th1 responses was observed in 4 of 8 animals in the 10 µg group and in 7 of 7 animals in the 100 µg group (one animal in this group could not be evaluated for technical reasons). Overall, Th1-directed CD4 T-cell response levels were highest in the 100 µg group but were measured in both mRNA-1273 groups (Figure 3A) and Th2 responses were low to undetectable (Figure 3B). CD8 T-cell responses were low to undetectable following mRNA-1273 vaccination (data not shown). CD40L is a cell-surface marker expressed after CD4 T-cell activation and mediates B-cell activation for efficient isotype switching. The results showed that 3 of 8 rhesus macaques in the 10 µg group and 7 of 7 rhesus macaques in the 100 µg group had S-specific CD40L and CD4 T-cell responses, and 100 µg of mRNA-1273 induced greater responses in rhesus macaques than the PBS control (Figure 3C). Nucleoside-modified RNA vaccines that encode various viral antigens have been shown to induce robust antibody responses in rhesus macaques and are associated with increased CD4 Tfh cells (Rogers et al 2020). Therefore, IL-21 (the canonical cytokine produced by Tfh cells) was measured. Four of 8 rhesus macaques in the 10 µg group and 7 of 7 rhesus macaques in the 100 µg group had IL-21 responses. The magnitude of responses was significantly greater in the 100 µg group than in either the PBS or 10 µg group, and the response in the 10 µg group was significantly greater than that in the PBS group (Figure 3D).

These data demonstrate that mRNA-1273 induced Th1-producing and IL-21-producing Tfh-cell responses with no evidence of Th2-directed CD4 T-cell responses.

Figure 3: T-Cell Responses Following mRNA-1273 Immunization in Rhesus Macaques

Abbreviations: PBS = phosphate-buffered saline; Tfh = T follicular helper; Th1 = T-helper 1; Th2 = T-helper 2.

Note: Open symbols represent animals with a probable nonresponse, and solid symbols represent animals with a probable response.

Source: [Corbett et al 2020](#), [Figure 2](#).

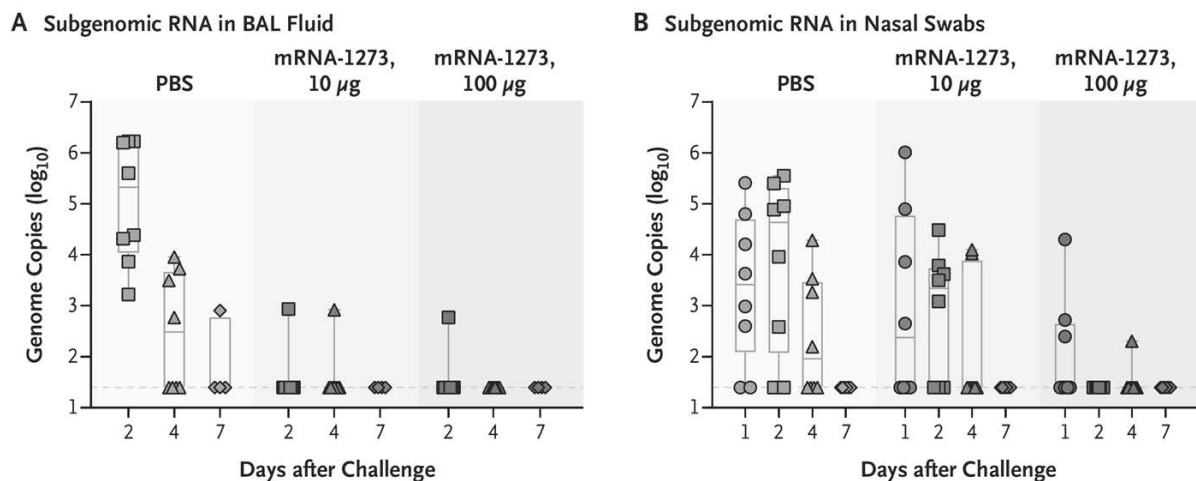
3.3. Protective Efficacy Against Upper and Lower SARS-CoV-2 Infection

In order to evaluate the protective efficacy of mRNA-1273, all animals were challenged with a total dose of 7.6×10^5 PFU of SARS-CoV-2, approximately equivalent to 10^6 50% tissue-culture infectious doses, 4 weeks post-boost by combined IT and IN routes. The route and dose for this challenge experiment were based on previous work challenging naïve nonhuman primates (IT/IN or endobronchially) with different doses of SARS-CoV-2 ([Appendix 2](#)). Based on what is known about viral levels in NS samples from humans following SARS-CoV-2 infection ([Zhou et al 2020](#)), the objective of this experiment was to deliver virus to both the upper and lower airways in order to detect levels of virus post-challenge that were similar to those in infected humans.

Two days after challenge, only 1 of 8 animals in both the 10 and 100 µg of mRNA-1273 groups had detectable sgRNA in BAL fluid ([Figure 4A](#)), compared with 8 of 8 animals in the PBS group. In NS samples ([Figure 4B](#)), 0 of 8 animals in the 100 µg group had detectable sgRNA by

Day 2, compared with 5 of 8 animals in the 10 μg group and 6 of 8 animals in the PBS control group. By Day 4, 1 of 8 animals in the 100 μg group and 2 of 8 animals in the 10 μg group had low levels of sgRNA RNA detected in NS samples.

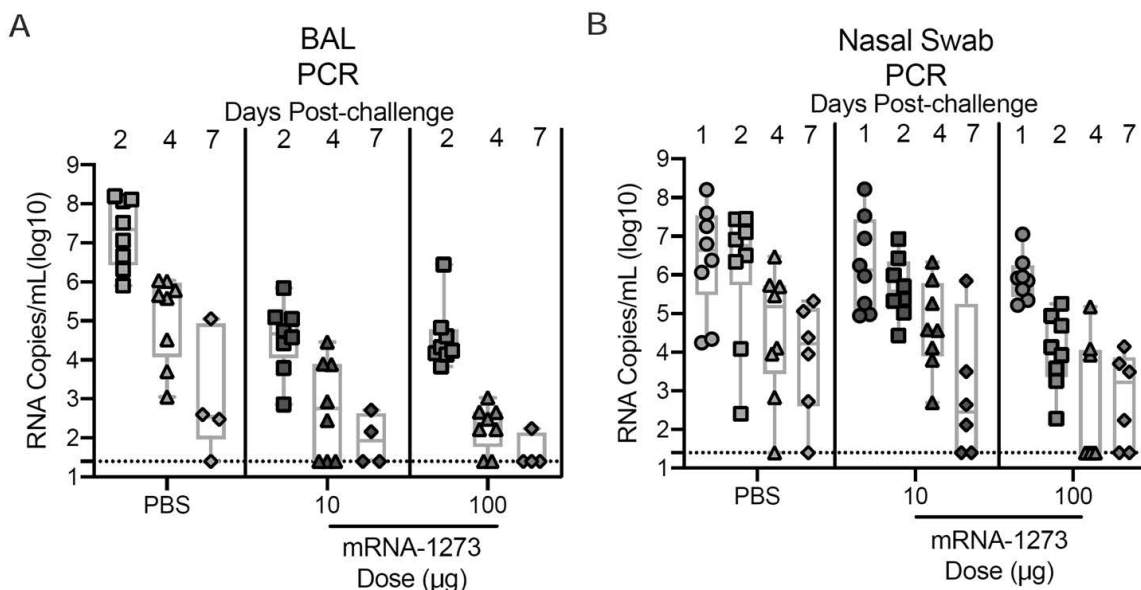
Figure 4: Efficacy of mRNA-1273 Against Upper and Lower Respiratory Viral Replication



Abbreviations: BAL = bronchoalveolar lavage; PBS = phosphate-buffered saline.

Source: [Corbett et al 2020, Figure 3](#)

Quantitation of total viral RNA by RT-qPCR revealed a decrease in BAL fluid for both the 10 and 100 μg groups by Day 2 post-challenge ([Figure 5A](#)), which was also observed in NS samples at the 100 μg dose ([Figure 5B](#)). By Day 4 post-challenge, rhesus macaques immunized with 10 μg of mRNA-1273 exhibited decreased virus in NS samples ([Figure 5B](#)). Peak levels of sgRNA on Days 2 through 7 (in BAL fluid and NS specimens) and total viral RNA (in BAL fluid) were lower in both groups vaccinated with mRNA-1273 than in the PBS-treated group. Animals vaccinated with 100 μg of mRNA-1273 had lower total viral RNA levels in NS samples than animals administered 10 μg of mRNA-1273 or PBS control.

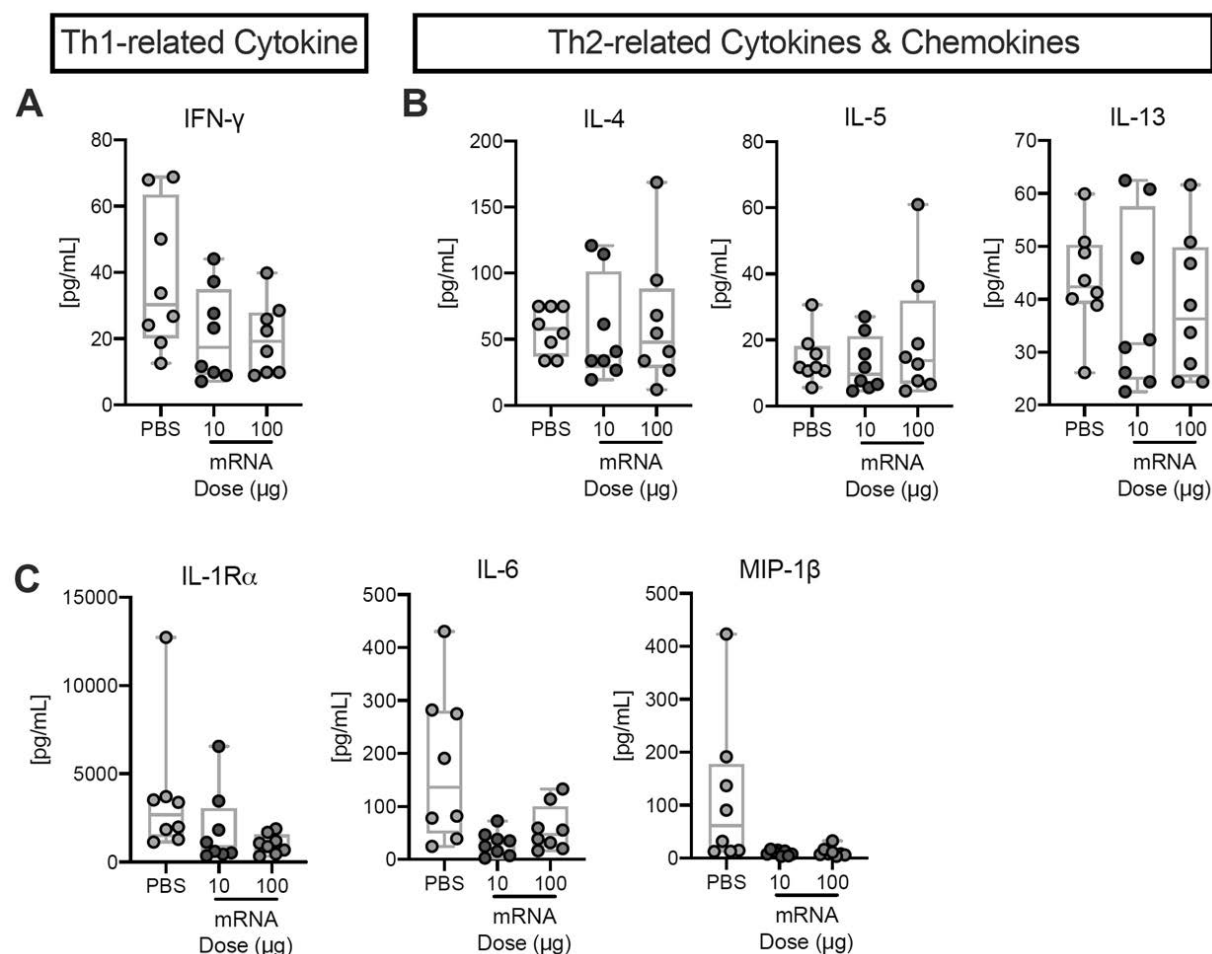
Figure 5: Efficacy of mRNA-1273 Against Upper and Lower Respiratory Viral Load

Abbreviations: BAL = bronchoalveolar lavage; PBS = phosphate-buffered saline; PCR = polymerase chain reaction

Source: [Corbett et al 2020, Supplemental Figure 3](#)

A panel of innate immune system cytokines and chemokines were assessed in BAL fluid at Day 2 and Day 4 post-challenge in order to further investigate viral infection in the lung. Within 2 days, there was limited inflammatory cytokine induction in the upper and lower airway of rhesus macaques in both groups, suggesting that the rapid control of virus was sufficient to limit innate immune activation ([Figure 6](#)).

Figure 6: Post-challenge BAL Cytokine and Chemokine Responses in mRNA-1273-Immunized Rhesus Macaques

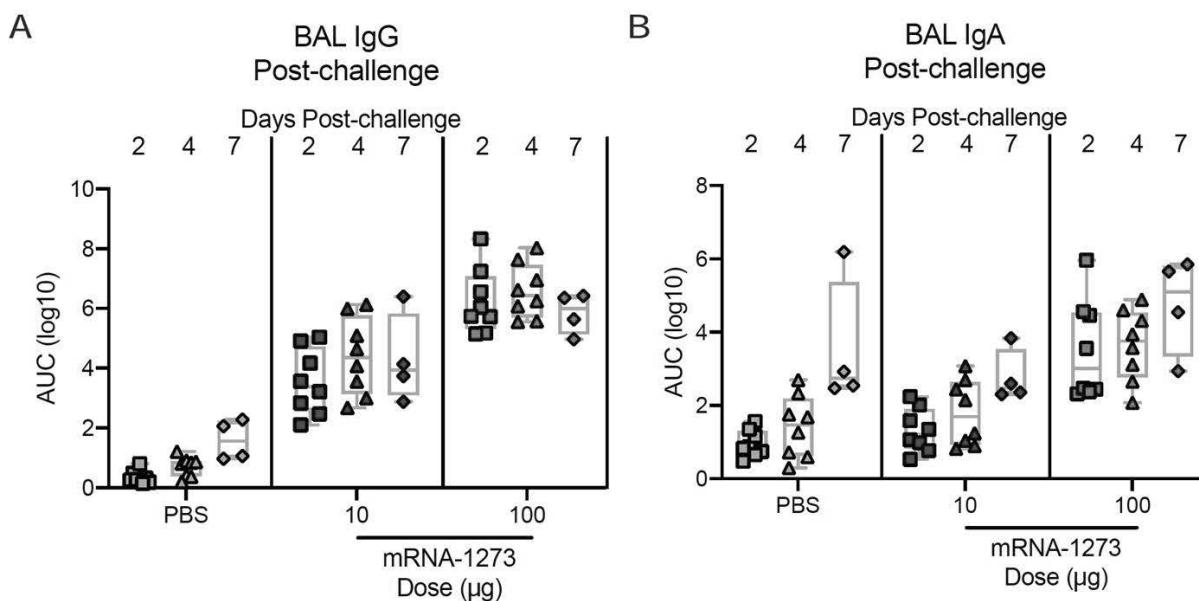


Abbreviations: BAL = bronchoalveolar lavage; IFN = interferon; IL = interleukin; MIP = macrophage inflammatory protein; mRNA = messenger RNA; PBS = phosphate-buffered saline; Th1 = T-helper 1; Th2 = T-helper 2.

Source: [Corbett et al 2020, Supplemental Figure 4](#)

Post-challenge antibody levels in BAL fluid were measured to explore potential immune mechanisms that may mediate rapid control of viral replication in the lung. A dose-dependent increase in S-specific IgG antibody titers was observed in BAL fluid of mRNA-1273-vaccinated rhesus macaques compared with levels in PBS control group animals. S-specific IgA responses in BAL fluid were lower than the IgG responses, and the IgA responses were higher in the 100 μ g group than the 10 μ g and control group ([Figure 7](#)).

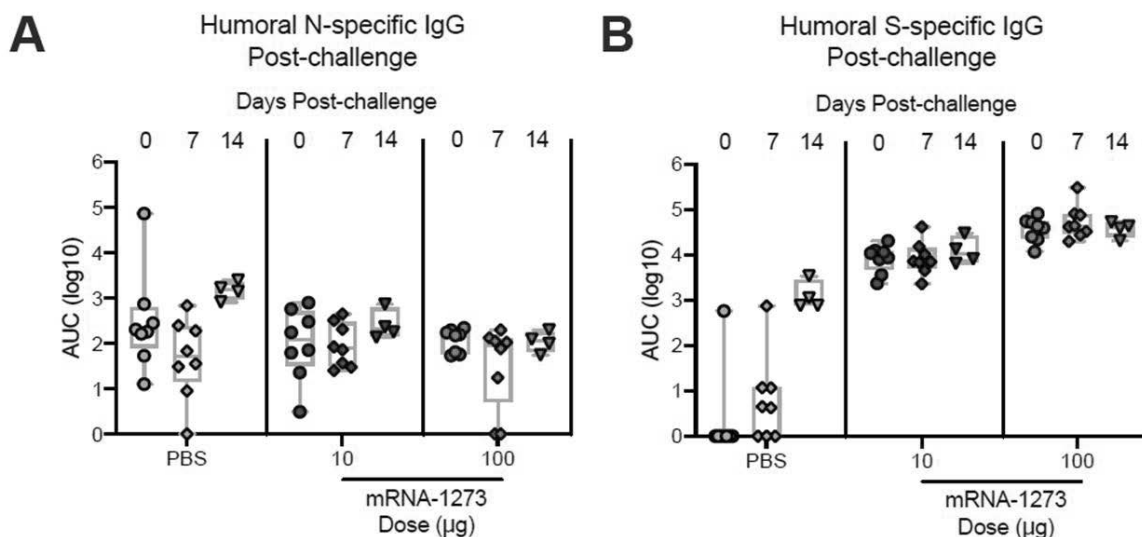
Figure 7: Post-challenge BAL Antibody Response in mRNA-1273-Immunized Rhesus Macaques



Abbreviations: AUC = area under the curve; BAL = bronchoalveolar lavage; Ig = immunoglobulin; PBS = phosphate-buffered saline.

Source: [Corbett et al 2020, Supplemental Figure 6](#)

Post-challenge humoral S- and N-specific IgG increased in control animals within 2 weeks post-challenge, whereas antibody levels in mRNA-1273-vaccinated animals remained high ([Figure 8](#)). The increase in control animals is indicative of an immune response to the challenge virus. In vaccinated animals, no anamnestic response was observed post-challenge, which further demonstrates that the animals were protected from viral challenge.

Figure 8 Post-challenge Humoral S-Specific and N-Specific IgG

Abbreviations: AUC = area under the curve; IgG = immunoglobulin G; PBS = phosphate-buffered saline.

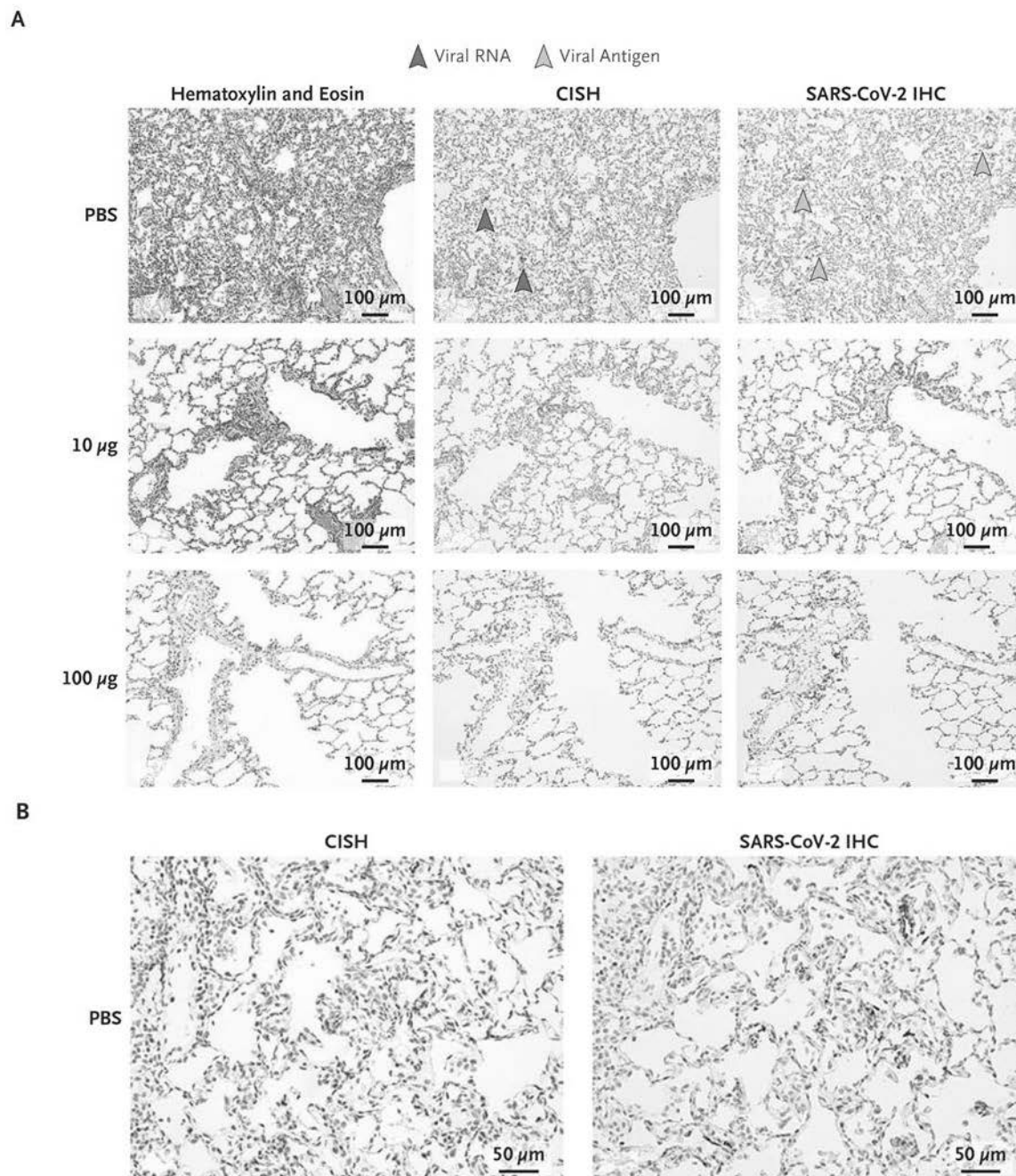
Source: [Corbett et al 2020, Supplemental Figure 7](#)

3.4. Pathology and Viral Load in Lung Tissue Post-Challenge

Seven or 8 days after challenge, lungs were harvested from 2 animals per group for histopathologic analysis and assessment of evidence of viral infection. SARS-CoV-2 infection in PBS control rhesus macaques caused moderate to severe inflammation that often involved the small airways and the adjacent alveolar interstitia, consistent with previous reports ([Chandrashekar et al 2020](#); [Munster et al 2020](#); [Rockx et al 2020](#)). Alveolar air spaces occasionally contained inflammatory cell infiltrates, alveolar capillary septa were moderately thickened, and moderate and diffuse type II pneumocyte hyperplasia was observed. Multiple pneumocytes were present in the lung sections from the PBS control group and were positive for both SARS-CoV-2 viral RNA and antigen, as assessed by cISH and immunohistochemistry ([Figure 9A](#) and [Figure 9B](#), [Table 3](#)). At Day 7 in rhesus macaques immunized with 10 µg of mRNA-1273, inflammation was mild and viral RNA was not detected ([Figure 9A](#), [Table 3](#)); however, at Day 8, one animal in the 10 µg group exhibited a single pneumocyte that was positive for viral antigen ([Table 3](#)). No significant inflammation was observed in the lungs of rhesus macaques immunized with 100 µg of mRNA-1273, and no viral RNA or antigen was detected at Day 7 or 8 post-challenge ([Figure 9A](#), [Figure 10](#), [Table 3](#)). In addition, lung sections from animals that were sacrificed at Day 14 or 15 after challenge showed no evidence of substantial inflammation and neither viral RNA nor viral antigen were detected in any of the

groups, including the control group. Vaccine-associated immunopathologic changes were not observed in any of the sections examined.

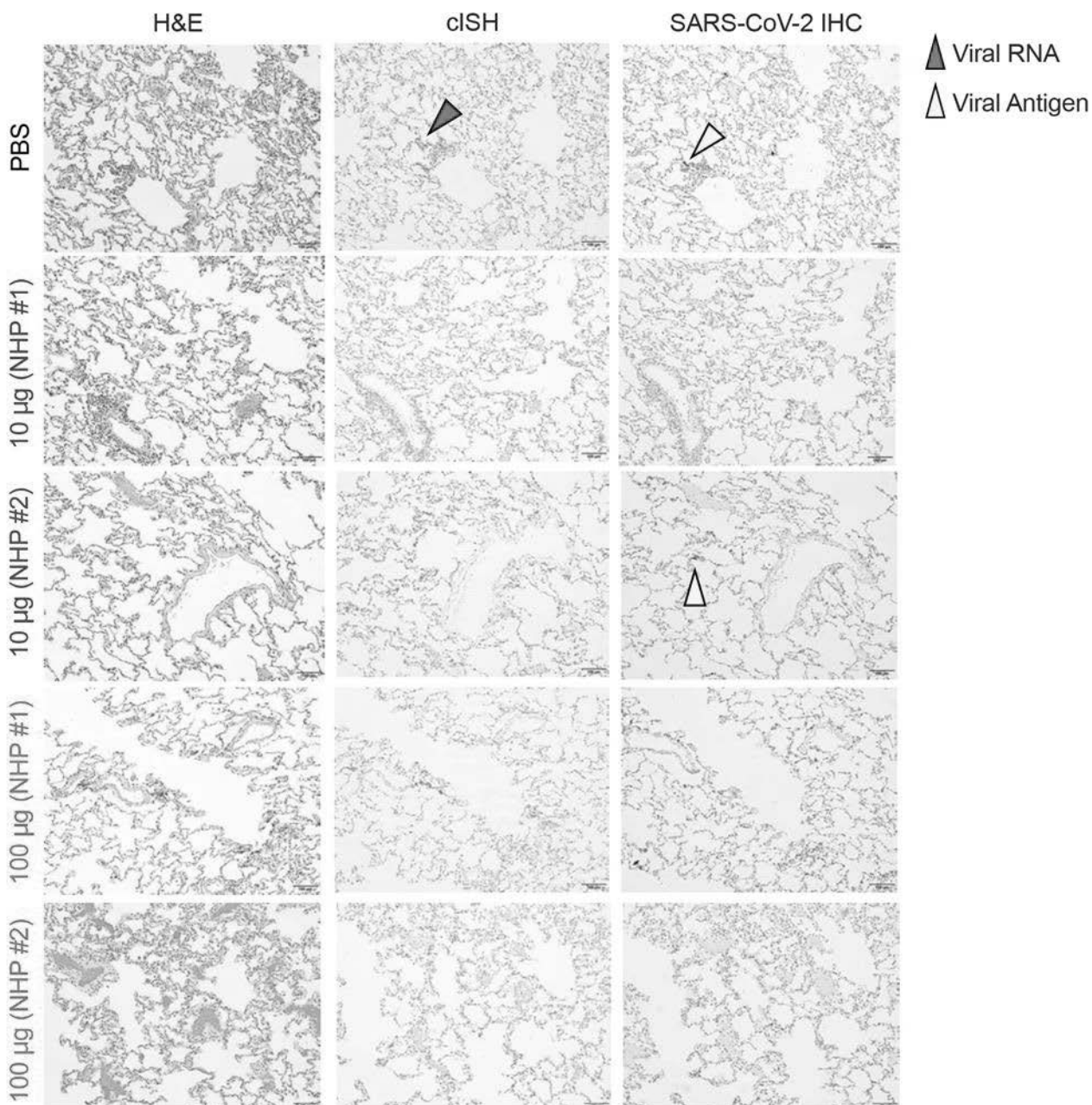
Figure 9: Lung Histopathology and Viral Detection 7 Days Post-challenge in mRNA-1273-Immunized Rhesus Macaques



Abbreviations: CISH = chromogenic in situ hybridization; IHC = immunohistochemistry; PBS = phosphate-buffered saline; SARS-CoV-2 = 2019 novel coronavirus.

Source: [Corbett et al 2020, Figure 4](#)

Figure 10: Lung Histopathology and Viral Detection 8 Days Post-challenge in mRNA-1273-Immunized Rhesus Macaques



Abbreviations: cISH = chromogenic in situ hybridization; H&E = hematoxylin and eosin; IHC = immunohistochemistry; NHP = non-human primate; PBS = phosphate-buffered saline; SARS-CoV-2 = 2019 novel coronavirus.

Source: [Corbett et al 2020, Supplemental Figure 8](#)

Table 3: Assessment of Lung Inflammation, Viral RNA, and Viral Antigen Following SARS-CoV-2 Challenge of mRNA-1273-Immunized Rhesus Macaques

Day Post-Challenge	mRNA Dose (µg)	Inflammation	ISH (RNA; R ^a /L ^b) ^c	IHC2 (SARS-CoV-2 Ag; R ^a /L ^b) ^c
7	PBS	Moderate	+/- / +/- ^c	+ / +
7	PBS	Moderate-severe	+ / +	+ / +
7	10	Minimal-mild	- / -	- / -
7	10	Mild-moderate	- / -	- / -
7	100	Minimal-absent	- / -	- / -
7	100	Minimal-absent	- / -	- / -
8	PBS	Moderate-severe	+/- / -	+ / -
8	PBS	Moderate	+/- / +	+ / +
8	10	Minimal-mild	- / -	- / -
8	10	Minimal-mild	- / -	^d +/- / -
8	100	Minimal-mild	- / -	- / -
8	100	Mild-moderate	- / -	- / -

Abbreviations: IHC = immunohistochemistry; ISH = in situ hybridization; mRNA = messenger RNA; PBS = phosphate-buffered saline; SARS-CoV-2 = 2019 novel coronavirus.

Note: Lung tissue was evaluated for presence of inflammation, SARS-CoV-2 viral RNA, and viral antigen.

^a Lung right side.

^b Lung left side.

^c - = not detected; +/- = occasional positive foci; + = multiple positive foci.

^d Viral antigen detected in a single positive pneumocyte.

4. CONCLUSIONS

This study in rhesus macaques vaccinated with mRNA-1273 shows rapid prevention of viral replication in upper and lower airways after a high-dose challenge with SARS-CoV-2. The ability to limit viral replication in both the lower and upper airways has important implications for whether vaccine-induced neutralizing activity can prevent both disease and transmission by SARS-CoV-2.

When compared to human convalescent sera, mRNA-1273 elicited more potent RBD- and S1_NTD-specific serum antibody responses than natural infection. In the 100 µg group, inhibition of ACE-2 binding to RBD was significantly higher than inhibition in the human convalescent serum and PBS control groups, and 4 weeks post-boost, potent neutralizing capacity was observed using the live SARS-CoV-2 reporter virus and the pseudovirus neutralization assay in both the 10 µg and 100 µg groups.

Intracellular cytokine staining revealed that, overall, Th1-directed CD4 T-cell and IL-21 producing Tfh-cell response levels were driven by vaccination with mRNA-1273, with responses higher in the 100 µg group than in the 10 µg group. Th2-directed CD4 T-cell responses were low to undetectable in both vaccine groups.

Evaluation of the protective efficacy against upper and lower SARS-CoV-2 infection revealed a reduction in detectable sgRNA in BAL fluid (7 out of 8 animals in both vaccinated groups) and NS samples (all animals in 100 µg group) as early as 2 days post-challenge. Specifically, no viral replication was detectable in the nose of any of the 8 animals in the 100 µg group by Day 2 after challenge.

Additionally, RT-qPCR showed a decrease in total viral RNA in BAL fluid and NS samples 2 days post-challenge. There was limited inflammatory cytokine induction in the upper and lower airway of rhesus macaques within 2 days, suggesting that the rapid control of the virus was sufficient to limit innate immune activation. There was also a dose-dependent increase in S-specific IgG antibody titers in BAL fluid compared with the titers in control animals. S-specific IgA responses in BAL fluid were lower than the IgG responses, and the IgA responses were higher in the 100 µg group than the 10 µg and control group.

Histopathology conducted at 7 or 8 days post-challenge on animals immunized with 100 µg of mRNA-1273 revealed no significant inflammation in the lungs, and no viral RNA or antigen was observed in the lungs. Furthermore, there was no evidence of vaccine-associated immunopathology.

In conclusion, vaccination of rhesus macaques with mRNA-1273 induced robust SARS-CoV-2 binding antibodies and neutralizing activity and rapid protection in the upper and lower airways after challenge, with no pathologic changes observed in the lungs.

5. REFERENCES

Chandrashekar A, Liu J, Martinot AJ, McMahan K, Mercado NB, Peter L, et al. SARS-CoV-2 infection protects against rechallenge in rhesus macaques. *Science*. 2020;369(6506):812-7.

Corbett KS, Edwards D, Leist SR, Abiona OM, Boyoglu-Barnum S, Gillespie RA, et al. SARS-CoV-2 mRNA vaccine development enabled by prototype pathogen preparedness. *bioRxiv*. 2020;2020.06.11.145920. Epub 2020 Jun 11.

Donaldson MM, Kao SF, Foulds KE. OMIP-052: An 18-color panel for measuring Th1, Th2, Th17, and Tfh responses in rhesus macaques. *Cytometry A*. 2019;95(3):261-3.

Finak G, McDavid A, Chattopadhyay P, Dominguez M, De Rosa S, Roederer M, et al. Mixture models for single-cell assays with applications to vaccine studies. *Biostatistics*. 2013;15(1):87-101.

Grifoni A, Weiskopf D, Ramirez SI, Mateus J, Dan JM, Moderbacher CR, et al. Targets of T cell responses to SARS-CoV-2 coronavirus in humans with COVID-19 disease and unexposed individuals. *Cell*. 2020;181(7):1489-501.e15.

Hassett KJ, Benenato KE, Jacquinet E, Lee A, Woods A, Yuzhakov O, et al. Optimization of lipid nanoparticles for intramuscular administration of mRNA vaccines. *Mol Ther Nucleic Acids*. 2019;15:1-11.

Hou YJ, Okuda K, Edwards CE, Martinez DR, Asakura T, Dinnon KH 3rd, et al. SARS-CoV-2 reverse genetics reveals a variable infection gradient in the respiratory tract. *Cell*. 2020;182(2):429-46.e14.

Meckiff BJ, Ramírez-Suástegui C, Fajardo V, Chee SJ, Kusnadi A, Simon H, et al. Single-cell transcriptomic analysis of SARS-CoV-2 reactive CD4⁺ T cells. *bioRxiv*. 2020;2020.06.12.148916. Epub 2020 Jun 13.

Munster VJ, Feldmann F, Williamson BN, van Doremalen N, Pérez-Pérez L, Schulz J, et al. Respiratory disease in rhesus macaques inoculated with SARS-CoV-2. *Nature*. 2020. Epub 2020 May 12.

Neidleman J, Luo X, Frouard J, Xie G, Gill G, Stein ES, et al. SARS-CoV-2-specific T cells exhibit phenotypic features reflecting robust helper function, lack of terminal differentiation, and high proliferative potential. *bioRxiv*. 2020;2020.06.08.138826. Epub 2020 Aug 03.

Nelson J, Sorensen EW, Mintri S, Rabideau AE, Zheng W, Besin G, et al. Impact of mRNA chemistry and manufacturing process on innate immune activation. *Sci Adv*. 2020;6(26):eaaz6893.

Pallesen J, Wang N, Corbett KS, Wrapp D, Kirchdoerfer RN, Turner HL, et al. Immunogenicity and structures of a rationally designed prefusion MERS-CoV spike antigen. *Proc Natl Acad Sci U S A*. 2017;114(35):E7348-57.

Peng Y, Mentzer AJ, Liu G, Yao X, Yin Z, Dong D, et al. Broad and strong memory CD4⁺ and CD8⁺ T cells induced by SARS-CoV-2 in UK convalescent COVID-19 patients. *bioRxiv*. 2020;2020.06.05.134551. Epub 2020 Jun 08.

Rockx B, Kuiken T, Herfst S, Bestebroer T, Lamers MM, Oude Munnink BB, et al. Comparative pathogenesis of COVID-19, MERS, and SARS in a nonhuman primate model. *Science*. 2020;368(6494):1012-5.

Rogers TF, Zhao F, Huang D, Beutler N, Burns A, He WT, et al. Isolation of potent SARS-CoV-2 neutralizing antibodies and protection from disease in a small animal model. *Science*. 2020;369(6506):956-63.

Scobey T, Yount BL, Sims AC, Donaldson EF, Agnihothram SS, Menachery VD, et al. Reverse genetics with a full-length infectious cDNA of the Middle East respiratory syndrome coronavirus. *Proc Natl Acad Sci U S A*. 2013;110(40):16157-62.

Sekine T, Perez-Potti A, Rivera-Ballesteros O, Strålin K, Gorin JB, Olsson A, et al. Robust T cell immunity in convalescent individuals with asymptomatic or mild COVID-19. *bioRxiv*. 2020;2020.06.29.174888. Epub 2020 Jun 29.

van Doremalen N, Lambe T, Spencer A, Belij-Rammerstorfer A, Purushotham JN, Port JR, et al. ChAdOx1 nCoV-19 vaccination prevents SARS-CoV-2 pneumonia in rhesus macaques. *bioRxiv*. 2020;2020.05.13.093195. Epub 2020 May 13.

Wang F, Flanagan J, Su N, Wang LC, Bui S, Nielson A, et al. RNAscope: a novel in situ RNA analysis platform for formalin-fixed, paraffin-embedded tissues. *J Mol Diagn*. 2012;14(1):22-9.

Weiskopf D, Schmitz KS, Raadsen MP, Grifoni A, Okba NM, Endeman H, et al. Phenotype and kinetics of SARS-CoV-2-specific T cells in COVID-19 patients with acute respiratory distress syndrome. *Sci Immunol*. 2020;5(48):eabd2071.

Wölfel R, Corman VM, Guggemos W, Seilmaier M, Zange S, Müller MA, et al. Virological assessment of hospitalized patients with COVID-2019. *Nature*. 2020;581(7809):465-9.

Wrapp D, Wang N, Corbett KS, Goldsmith JA, Hsieh CL, Abiona O, et al. Cryo-EM structure of the 2019-nCoV spike in the prefusion conformation. *Science*. 2020;367(6483):1260-3.

Yount B, Curtis KM, Fritz EA, Hensley LE, Jahrling PB, Prentice E, et al. Reverse genetics with a full-length infectious cDNA of severe acute respiratory syndrome coronavirus. *Proc Natl Acad Sci U S A*. 2003;100(22):12995-3000.

Yu J, Tostanoski LH, Peter L, Mercado NB, McMahan K, Mahrokhian SH, et al. DNA vaccine protection against SARS-CoV-2 in rhesus macaques. *Science*. 2020;369(6505):806-11.

Zhou T, Teng IT, Olia AS, Cerutti G, Gorman J, Nazzari A, et al. Structure-based design with tag-based purification and in-process biotinylation enable streamlined development of SARS-CoV-2 spike molecular probes. *bioRxiv*. 2020:2020.06.22.166033. Epub 2020 Jun 23.

6. REPORT APPROVAL

Study Director:

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Viral Pathogenesis Laboratory, VRC, NIH
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Date

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Viral Pathogenesis Laboratory, VRC, NIH
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October 19, 2020

Date

7. APPENDICES

7.1. Appendix 1 Summary of Analysis: Lot AMPDP-200005 Release



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SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



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Test	Method	Target Attributes	Results
Lipid Identification:			
SM-102	DPTM-0019	(b) (4)	
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Content (mg/mL)			
SM-102	DPTM-0019	(b) (4)	(b) (4)
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Impurities (% Area)			
Individual Impurities	DPTM-0019	(b) (4)	
Total Impurities			
pH	USP <791> DPTM-0028		
Osmolality	USP <785> DPTM-0029		
Bacterial Endotoxin	USP <85>		
Bioburden	USP<61>		
Particulate Matter	USP <788> Method 2		



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REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval

7.2. Appendix 2 Supplemental Appendix 2 – Table S3 (Corbett et al 2020)

Worksheet: NHP model development data																		
						Protection Post-Challenge (viral RNA copies)												
Days:						1		2		3		4		7		14		
Group	Animal ID	Gender	Age at Week 0 (Years)	Weight (Kgs)	Dose (PFU)	Challenge Route	NS	BAL	NS	BAL	NS	BAL	NS	BAL	NS	BAL		
1	A12V195	F	7	8.5	1 x 10 ⁵	IT/IN	6580	N/C	267	N/C	25	N/C	-	-	25	N/C	25	N/C
	A12V035	F	8	9.0			108000	N/C	34900000	N/C	ND	N/C	-	-	337000	N/C	25	N/C
	A13V009	M	7	10.1			143000	N/C	69800000	N/C	1400000	N/C	-	-	2260000	N/C	8640	N/C
2	DFTM	M	7	10.7	1 x 10 ⁵	IT/IN	2870	N/C	229	450000000	219	N/C	-	-	-	-	-	-
	0C6	F	7	8.5			617000	N/C	945000	1140000	3720000	N/C	-	-	-	-	-	-
	A12V150	F	7	6.7			400000	N/C	9990000	5600000	292000	N/C	-	-	-	-	-	-
3	13N001	F	7	10.2	1 x 10 ⁵	EB	25	N/C	25	N/C	25	N/C	-	-	25	N/C	25	N/C
	59	F	7	6.6			25	N/C	25	N/C	25	N/C	-	-	25	N/C	25	N/C
	A12V165	M	7	14.3			25	N/C	25	N/C	25	N/C	-	-	25	N/C	25	N/C
4	A12V098	F	7	5.9	1 x 10 ⁵	IT/IN	974000	N/C	13000000	N/C	3220000	N/C	255000	320000	-	-	-	-
	13C210	M	6	7.8			1000000	N/C	41300000	N/C	34200000	N/C	750000	211000	-	-	-	-
	DF5Z	M	7	13.3			2330	N/C	444	N/C	223	N/C	224	N/A	-	-	-	-
5	09Z	F	7	6.9	1 x 10 ⁴	IT/IN	6690	1160000	25	N/C	25	N/C	-	-	-	-	-	-
	A12V038	M	8	10.8			25	1620000	71100000	N/C	51700000	N/C	-	-	-	-	-	-
	14C075	M	6	8.6			13700	91800	15900	N/C	1200000	N/C	-	-	-	-	-	-
6	13C373	M	6	8.6	1 x 10 ⁴	IT/IN	2120	N/C	26600	N/C	25	168000	-	-	25	N/C	-	-
	DF4K	M	7	11.6			4470	N/C	7730	N/C	2210	6340	-	-	3680	N/C	-	-
	A11V116	F	8	9.4			15800	N/C	105000	N/C	35500	2270000	-	-	7200	N/C	-	-
7	A12V093	M	7	12.6	1 x 10 ⁵	EB	25	N/C	25	37500000	25	N/C	-	-	-	-	-	-
	13N004	M	7	14.0			25	N/C	1330000	12400000	25	N/C	-	-	-	-	-	-
	A12V179	F	7	5.5			25	N/C	13300000	56800000	25	N/C	-	-	-	-	-	-
8	A12V039	M	8	12.9	1 x 10 ⁵	EB	25	N/C	241000	N/C	273000000	N/C	N/C	739000	-	-	-	-
	12N020	M	7	13.9			25	N/C	25	N/C	25	N/C	N/C	226000	-	-	-	-
	A12V097	F	7	7.0			253	N/C	25	N/C	727	N/C	N/C	29200	-	-	-	-
9	A12V185	M	7	11.3	1 x 10 ⁴	EB	25	4400000	25	N/C	25	-	-	-	-	-	-	-
	A12V115	F	7	7.8			25	4600000	704	N/C	25	-	-	-	-	-	-	-
	DFLI	M	7	7.5			25	1510000	25	N/C	25	-	-	-	-	-	-	-
10	A12V183	F	7	8.7	1 x 10 ⁴	EB	25	N/C	25	N/C	25	121000	-	-	-	-	-	-
	DFNK	M	7	13.5			25	N/C	25	N/C	25	249000	-	-	-	-	-	-
	DF9K	M	7	8.8			25	N/C	25	N/C	25	486000	-	-	-	-	-	-

25 = <50
(-) indicated sample not tested

25 = <50
(-) Indicated sample not tested